Project Topic: (Xinwen 1)

Optimizing Collective Escape in Predator–Prey Systems

Collective motion is common in prey species such as fish, birds, and insects, where coordinated group behavior can reduce predation risk. However, stronger coordination is not always better: while cohesive groups may transmit information quickly, they may also lose maneuverability or become easier targets. In this project, students will develop a predator–prey model in which prey follow Vicsek-inspired rules with alignment, cohesion, repulsion, limited perception, and predator avoidance, while predators may use different pursuit strategies [1, 2].

The main goal is to determine which collective movement strategies maximize prey survival. You will formulate your own optimization problem by varying behavioral parameters, group size, sensory information, environmental conditions, or predator strategies, and evaluate performance using measures such as survival probability, capture rate, response time, and group fragmentation.



Fig.1 Photo of gray shark surrounded with fishes Self-organized collective behavior is widespread in living systems, such as schools of fish. By coordinating their motion, fish can respond collectively to threats, improving the chances of avoiding predators such as gray sharks. *Photo by Tony Studio on* iStock

References:

[1] Vicsek, T., et al. *Novel Type of Phase Transition in a System of Self-Driven Particles*, Phys. Rev. Lett. **75**, 1226–1229 (1995).
https://doi.org/10.1103/PhysRevLett.75.1226

[2] Bartashevich, P., et al. *Collective anti-predator escape manoeuvres through optimal attack and avoidance strategies*, Commun. Biol. **7**, 1586 (2024).
https://doi.org/10.1038/s42003-024-07267-2